

Submission Report

1. Search formulation

State = grid coordinate; actions = Up/Down/Left/Right; transition = valid non-obstacle neighboring cell; initial state = S; goal = G; cost = 1 per move. The problem is deterministic.

2. Agent design

A* stores frontier nodes in a priority queue using $f(n)=g(n)+h(n)$, records parent pointers for reconstruction, and tracks expanded states.

3. Prompt used

Implement A* for the supplied warehouse grid. Use grid positions as states, four unit-cost moves, Manhattan distance as $h(n)$, a frontier, $g/h/f$, repeated-state avoidance, path reconstruction, and reporting of path length and states expanded. Also implement BFS for comparison.

4. Test results

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Original A*: found= True length= 40 expanded= 64
Original BFS: found= True length= 40 expanded= 64
Trivial A*: length= 1 expanded= 2
No-solution A*: found= False expanded= 9
Alternative paths A*: length= 4 expanded= 5
Manhattan found= True length= 40 expanded= 64
Zero found= True length= 40 expanded= 64
Euclidean found= True length= 40 expanded= 64
2x Manhattan found= True length= 40 expanded= 64
```

5. Interpretation

On this supplied map and implementation, BFS and A* both found a length-40 path and expanded 64 states. The heuristic variants tested also produced length 40 and 64 expansions. This is an observation for this experiment, not a general claim about the algorithms.